Name: _____

Instructions: All solutions should be prepared carefully and neatly. Any questions that ask you to explain or justify should have answers written in complete sentences. All solution sets shall be completed on this packet and handed in at the beginning of class on the due date below. Careless presentation (e.g. bad handwriting, pen scribbles, doodles, wasted space, etc) will result in a deduction of points at my discretion. If the question asks to submit a graph or plot, be sure to print the plot and include it in your final lab report. Use a PENCIL and if you make a mistake, use an eraser. Each student should prepare their own solutions that demonstrate their own comprehension of the material. Submitted work that does not demonstrate clearly the process by which one arrived at the answer will not receive credit of any kind. Academic dishonesty will not be tolerated.

Desmos Lab II

MATH 181 – Calculus I

Due: Friday, October 10

Read: Section 2.6 and 3.1

Problem Number	Available Points	Your Points
1	10	10
2	10	10
3	10	10
Total	30	30

1. Plotting a Function with its Tangent Line: In this part, we'll consider a basic function and derive the formula for the tangent line. Then, we'll generalize it so that Desmos plots the tangent line equation for any function at any point.

   (a) Open a new graph in Desmos and save it as "The Tangent Line."

   (b) Define the following function in Desmos: $f(x) = \frac{1}{3}x^3 + x^2 - 4x$. Graph this function on the window $x \in [-6, 6]$ and $y \in [-5, 20]$. Plot the curve in black with width size 4.

   (c) In the space below, find the value of $f(-2)$. Keep your answer in fraction form. In Desmos, plot this point on the curve, in orange, but let Desmos do the calculation, just type $(-2, f(-2))$ on a new line. Make it red with size 20.

   We have $f(-2) = \frac{1}{3}(-2)^3 + (-2)^2 - 4(-2) = -\frac{8}{3} + 4 + 8 = \frac{28}{3}$

   (d) In the space below, use the limit definition of the derivative at a point to calculate $f'(-2)$. Show all work.

   We have

$$\begin{aligned} f'(-2 &= \lim_{h\to 0} \frac{f(-2+h) - f(-2)}{h} \\ &= \lim_{h\to 0} \frac{\frac{1}{3}(-2+h)^3 + (-2+h)^2 - 4(-2+h) - \frac{28}{3}}{h} \\ &= \lim_{h\to 0} \frac{-\frac{8}{3} + 4h - 2h^2 + \frac{1}{3}h^3 + 4 - 4h + h^2 + 8 - 4h - \frac{28}{3}}{h} \\ &= \lim_{h\to 0} \frac{h\left(4 - 2h + \frac{1}{3}h^2 - 4 + h - 4\right)}{h} \\ &= \lim_{h\to 0} -4 - h + \frac{1}{3}h^2 \\ &= -4 \end{aligned}$$

   (e) Now that we have $f(-2)$ and $f'(-2)$, write down the equation of the tangent line in the space below. On a new line in Desmos, graph the tangent line in green with thickness 3.

   The general equation for the tangent line is

$$y - f(-2) = f'(-2)\big(x - (-2)\big) \quad \Rightarrow \quad y = f(-2) + f'(-2)(x+2),$$

   which, plugging in the appropriate values gives:

$$y = \frac{28}{3} - 4(x+2) \quad \Rightarrow \quad y = -4x + \frac{4}{3}.$$

   (f) Let's streamline this process! In the space below, write down the formula for the equation of the tangent line at *any* point $(c, f(c))$.

   In the equation above, we can replace every instance of "-2" with "c." This gives

$$y = f(c) + f'(c)(x - c).$$

   (g) In Desmos, on a new line, define $c = -2$. Then, change your point $(-2, f(-2))$ to $(c, f(c))$. Also, delete the line that held the equation of the tangent line, and replace it with your answer to part (f). You should see the same line you plotted earlier. Shorten the line by restricting the domain to $[c-2, c]$, so that it only has a tail. Push the play button for c. What happens?

   When we push play on c, we see that the tangent line "crawls" along the curve because the point at which that tangent line is drawn changes, but Desmos automatically graphs the new tangent line at each point.

(h) Now, let's make a train by defining the list `t = [c-1, c-.9, ... c-.1]` on a new line in a folder called "Train." Define the set of points $(t, f(t))$ on a new line. On another line, type the equation of the tangent line again, but evaluate it at t. Restrict the domain to be between $[t-2, t]$. Push play on c.

(i) Play with the train by changing the function or lengthening the tail. When you submit this assignment on D2L, share the url to this graph.

https://www.desmos.com/calculator/xx6wgmmlq8

2. Continuity and Differentiability: We consider a piecewise function and try to determine whether it is continuous and/or differentiable. Then we change the function to make it both.

   (a) Open a new graph in Desmos and save it as "Continuous and Differentiable."

   (b) Define the following function in Desmos:

$$f(x) = \begin{cases} x^2 & x \leq 1 \\ -3x + 1 & x > 1 \end{cases}$$

   Make the window $x \in [-3, 3]$ and $y \in [-4, 3]$. Plot the curve in black with thickness 5. Make sure to plot an open circle and a closed circle where appropriate.

   (c) Is this function continuous at $x = 1$? Is it differentiable at $x = 1$? Justify your answer to the second question by explicitly computing right and left hand limits.

   To check continuity, we check the following three things: $f(1)$, $\lim_{x\to 1^-} f(x)$, and $\lim_{x\to 1^+} f(x)$. It follows quickly that $f(1) = 1^2 = 1$. As for the limits, we have

$$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} x^2 = 1^2 = 1,$$

   and

$$\lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} -3x + 1 = -3(1) + 1 = -3 + 1 = -2.$$

   Since all three of these values are not equal, it follows that the function is not continuous at $x = 1$. Therefore, there is no w ay the function is differentiable at $x = 1$. We'll show it using limits. Consider the slope on the left side of $x = -1$:

$$\begin{aligned} \lim_{x\to 1^-} \frac{f(x) - f(1)}{x - 1} &= \lim_{x\to 1^-} \frac{x^2 - 1}{x - 1} \\ &= \lim_{x\to 1^-} \frac{(x-1)(x+1)}{x - 1} \\ &= \lim_{x\to 1^-} x + 1 \\ &= 1 + 1 \\ &= 2 \end{aligned}$$

   The derivative on the right hand side is

$$\begin{aligned} \lim_{x\to 1^+} \frac{f(x) - f(1)}{x - 1} &= \lim_{x\to 1^+} \frac{-3x + 1 - 1}{x - 1} \\ &= \lim_{x\to 1^+} \frac{-3x}{x - 1} \\ &= -\infty. \end{aligned}$$

   Therefore, the derivative at $x = 1$ does not exist, and so the function is not differentiable at $x = 1$.

   (d) Type $f'(1)$ into a new line in Desmos. Did it give an answer? Is the answer correct? Explain.

   When we type $f'(1)$ into Desmos, it does give an answer of $f'(1) = 2$, but this is incorrect. We notice that it is reporting the derivative on the left hand side of the point at $x = 1$ because that is the side in which the "$\leq$" is located when defining the piecewise function. This is essentially not how we take derivatives using the limit definition, so Desmos is wrong.

(e) Let's "fix" this function. Instead of defining the function to the right side of $x = 1$ as $-3x + 1$, define it as $ax + b$, where a and b are sliders. Find a set of values for a and b that make the function continuous, but not differentiable. Write them below.

Answers vary, but one combination that makes this function continuous but not differentiable would be $a = 1$ and $b = 0$.

(f) Now, find a set of values for a and b that makes the function both continuous and differentiable. Write them below. Using these values for a and b, show using one-sided limits that the function is indeed differentiable at $x = 1$.

There is only one answer, which is $a = 2$ and $b = -1$. The left hand limit for the derivative at $x = 1$ will remain the same since the left function does not depend on a and b. The right hand limit becomes

$$\begin{aligned}\lim_{x\to 1^+} \frac{f(x) - f(1)}{x - 1} &= \lim_{x\to 1^+} \frac{2x - 1 - 1}{x - 1} \\ &= \lim_{x\to 1^+} \frac{2x - 2}{x - 1} \\ &= \lim_{x\to 1^+} \frac{2(x - 1)}{x - 1} \\ &= \lim_{x\to 1^+} 2 \\ &= 2\end{aligned}$$

Therefore, the function is now differentiable at $x = 1$ since both left and right side limits of the derivatives equal 2.

(g) Is there any way the function can be differentiable but not continuous? Explain.

This is impossible because if a function is differentiable, then the function value exists at the point and the values of the function near that point must be getting closer and closer together. This essentially implies continuity. It is more difficult for a function to be differentiable than continuous; thus, if a function is differentiable, it brings continuity with it.

(h) When you submit this assignment on D2L, share the url to this graph.

`https://www.desmos.com/calculator/cuqt2ghyhi`

3. What does the Derivative Function Tell Us?: In this part, we consider two polynomial functions and try to determine what information about the original function the first and second derivatives tell us about it. We can deduce certain characteristics about a function by inspecting the graph of the derivatives.

   (a) Open a new graph in Desmos and save it as "What the Derivative Tells Us."

   (b) On a new line, define the function $f(x) = x^3 - 3x^2 + 2$. Graph it on an appropriate window so you can see its characteristics nicely.

   (c) On the line below it, type $y = f'(x)$. What do you see?
   Even though Desmos does not find the explicit form of the derivative function, it does graph the derivative function for us, which we can see is a parabolic looking function.

   (d) By inspecting the graph of $y = f'(x)$ come up with the actual function. Write it down below. (You could also use the limit definition.)
   I see that the zeros are $x = 0$ and $x = 2$, and the y-intercept is $f'(0) = 0$. Therefore, our function is $f'(x) = ax(x-2)$, for some unknown value of a. Since we can see that $f'(1) = -3$, we have $f'(1) = a(1)(1-2) = -a$. Therefore $a = 3$. Finally, our function is $f'(x) = 3x(x-2) = 3x^2 - 6x$.

   (e) Find the zeros of $f'(x)$, either by inspecting the graph of $f'(x)$ or algebraically. Write them below.
   By inspection, we see that the zeros of $f'(x)$ are $x = 0$ and $x = 2$.

   (f) What is true about the graph of $f(x)$ at the zeros of $f'(x)$?
   At the zeros of $f'(x)$, we see that the original function $f(x)$ has a relative maximum at $x = 0$ and a relative minimum at $x = 2$. Also, we can see that the original function $f(x)$ has a zero slope at those points, which makes sense since the derivative (or slope) has an x-intercept there.

   (g) Do a sign analysis by writing a number line below, indicating where $f'(x) > 0$ and where $f'(x) < 0$. What is happening on the graph of $f(x)$ on these intervals?
   By inspection, we can see that $f'(x) > 0$ on the interval $(-\infty, 0) \cup (0, \infty)$ and $f'(x) < 0$ on $(0, 2)$. On the same interval where $f'(x) > 0$, we can see that is where $f(x)$ is increasing. Likewise, on the interval where $f'(x) < 0$, we can see that the original graph of $f(x)$ is decreasing.

   (h) Now graph $y = f''(x)$. What type of function is it? Derive a formula for $f''(x)$ and write it below.
   We can see that the second derivative is a line with y-intercept at $b = -6$ and slope of $m = 6$. Therefore, we the equation for the second derivative is $f''(x) = 6x - 6$.

   (i) What is true about $f'(x)$ at the zeros of $f''(x)$? What is true about $f(x)$ at the zeros of $f''(x)$?
   There is a sole zero of $f''(x)$ at $x = 1$. The graph of $f'(x)$ has a relative minimum at $x = 1$. The graph of the original function $f(x)$ is decreasing the most at the point $x = 1$, which indicates that it has an inflection point there. (There is also a root of $f(x)$ at $x = 1$, but this is a coincidence and $f''(x)$ has no relation to this fact.)

   (j) Find where $f''(x) > 0$ and where $f''(x) < 0$. Describe what is happening with $f'(x)$ and $f(x)$ on these intervals.
   We see that $f''(x) > 0$ on $(1, \infty)$. The first derivative is increasing on this same interval and the original function is concave up on this same interval. We also see that $f''(x) < 0$ on $(-\infty, 1)$. The first derivative is decreasing on this same interval and the original function is concave down on this same interval.

   (k) What is $f'''(x)$? What is $f^{(4)}(x)$?
   After graphing each function, we can see that $f'''(x) = 6$ and $f^{(4)}(x) = 0$. They are both constant functions.

   (l) Explore these properties with the function $f(x) = \sin(x)$. Do similar properties of $f(x)$, $f'(x)$, and $f''(x)$ exist? Explain.
   Answers vary, but the same correspondence between increasing/decreasing occurs for the $\sin(x)$ function and it's derivatives.

(m) When you submit this assignment on D2L, share the url to this graph.
https://www.desmos.com/calculator/erogvf401k